一维高斯积分：

Points	Locations	Weights	Order
1	0.	2.	1
2	−0.57735 02691 89626 0.57735 02691 89626	1. 1.	3
3	−0.77459 66692 41483 0. 0.77459 66692 41483	0.55555 55555 55556 0.88888 88888 88889 0.55555 55555 55556	5
4	−0.86113 63115 94053 −0.33998 10435 84856 0.33998 10435 84856 0.86113 63115 94053	0.34785 48451 37454 0.65214 51548 62546 0.65214 51548 62546 0.34785 48451 37454	7
5	−0.90617 98459 38664 −0.53846 93101 05683 0. 0.53846 93101 05683 0.90617 98459 38664	0.23692 68850 56189 0.47862 86704 99366 0.56888 88888 88889 0.47862 86704 99366 0.23692 68850 56189	9
6	−0.93246 95142 03152 −0.66120 93864 66265 −0.23861 91860 83197 0.23861 91860 83197 0.66120 93864 66265 0.93246 95142 03152	0.17132 44923 7917 0.36076 15730 48139 0.46791 39345 72691 0.46791 39345 72691 0.36076 15730 48139 0.17132 44923 7917	11

Example 1：

Evaluate the following integral using two, three and four point Gauss quadrature.

$$I = \int_{-1}^{1} (0.2 + 25\,s - 200\,s^2 + 675\,s^3 - 900\,s^4 + 400\,s^5)\,ds$$

$$f(s) = 400\,s^5 - 900\,s^4 + 675\,s^3 - 200\,s^2 + 25\,s + 0.2$$

Using two point formula the integral is evaluated as follows.

i	s_i	$f(s_i)$	w_i	$w_i f(s_i)$
1	−0.57735	−336.464	1.	−336.464
2	0.57735	3.53091	1.	3.53091
				$I \approx -332.933$

Using three point formula the integral is evaluated as follows.

i	s_i	$f(s_i)$	w_i	$w_i f(s_i)$
1	−0.774597	−888.418	0.555556	−493.566
2	0.	0.2	0.888889	0.177778
3	0.774597	0.818488	0.555556	0.454716
				$I \approx -492.933$

Since the given function f is a 5th order polynomial, the three point formula should give us the exact integral. A direct evaluation of the integral confirms this. As the following computation with four points demonstrates using any formula with $n \geq 3$ will give the exact integral for this function.

i	s_i	$f(s_i)$	w_i	$w_i f(s_i)$
1	−0.861136	−1285.01	0.347855	−446.998
2	−0.339981	−71.784	0.652145	−46.8136
3	0.339981	1.90044	0.652145	1.23936
4	0.861136	−1.03804	0.347855	−0.361088
				$I \approx -492.933$

二维高斯积分：

Quadrature	Points	s_I	t_J	$w_I \times w_J$
1×1	1	0.	0.	4.
2×2	1	−0.57735	−0.57735	1.
	2	−0.57735	0.57735	1.
	3	0.57735	−0.57735	1.
	4	0.57735	0.57735	1.
3×3	1	−0.774597	−0.774597	0.308642
	2	−0.774597	0.	0.493827
	3	−0.774597	0.774597	0.308642
	4	0.	−0.774597	0.493827
	5	0.	0.	0.790123
	6	0.	0.774597	0.493827
	7	0.774597	−0.774597	0.308642
	8	0.774597	0.	0.493827
	9	0.774597	0.774597	0.308642
4×4	1	−0.861136	−0.861136	0.121003
	2	−0.861136	−0.339981	0.226852
	3	−0.861136	0.339981	0.226852
	4	−0.861136	0.861136	0.121003
	5	−0.339981	−0.861136	0.226852
	6	−0.339981	−0.339981	0.425293
	7	−0.339981	0.339981	0.425293
	8	−0.339981	0.861136	0.226852
	9	0.339981	−0.861136	0.226852
	10	0.339981	−0.339981	0.425293
	11	0.339981	0.339981	0.425293
	12	0.339981	0.861136	0.226852
	13	0.861136	−0.861136	0.121003
	14	0.861136	−0.339981	0.226852
	15	0.861136	0.339981	0.226852
	16	0.861136	0.861136	0.121003

Example 2：

Evaluate the following integral using 2×2, 3×2, and 3×3 point Gauss quadrature.

$$I = \int_{-1}^{1}\int_{-1}^{1}(400\,s^5 + 675\,s^3 + 25\,s - 900\,s^2\,t^6 - 200\,t^2 + 0.2)\,ds\,dt$$

$$f(s,t) = -900\,s^2\,t^6 - 200\,t^2 + 400\,s^5 + 675\,s^3 + 25\,s + 0.2$$

Using a 2×2 Gauss quadrature, the integral is computed as follows.

Point	s_i, t_j	$f(s_i, t_j)$	$w_i \times w_j$	$w_i \times w_j f(s_i, t_j)$
1	−0.57735 −0.57735	−247.575	1.	−247.575
2	−0.57735 0.57735	−247.575	1.	−247.575
3	0.57735 −0.57735	92.4198	1.	92.4198
4	0.57735 0.57735	92.4198	1.	92.4198

$$I \approx -310.311$$

With 3×2 Gauss quadrature, there are three points in the s directions and 2 points in the t direction as follows.

s direction points, s_i = {−0.774597, 0., 0.774597} weights, w_i = {0.555556, 0.888889, 0.555556}

t direction points, t_j = {−0.57735, 0.57735} weights, w_j = {1., 1.}

Thus the integral is computed as follows.

Point	s_i, t_j	$f(s_i, t_j)$	$w_i \times w_j$	$w_i \times w_j f(s_i, t_j)$
1	−0.774597 −0.57735	−531.085	0.555556	−295.047
2	−0.774597 0.57735	−531.085	0.555556	−295.047
3	0. −0.57735	−66.4667	0.888889	−59.0815
4	0. 0.57735	−66.4667	0.888889	−59.0815
5	0.774597 −0.57735	358.152	0.555556	198.973
6	0.774597 0.57735	358.152	0.555556	198.973

$$I \approx -310.311$$

Using a 3×3 Gauss quadrature, the integral is computed as follows.

Point	s_i, t_j	$f(s_i, t_j)$	$w_i \times w_j$	$w_i \times w_j f(s_i, t_j)$
1	−0.774597 −0.774597	−681.058	0.308642	−210.203
2	−0.774597 0.	−444.418	0.493827	−219.466
3	−0.774597 0.774597	−681.058	0.308642	−210.203
4	0. −0.774597	−119.8	0.493827	−59.1605
5	0. 0.	0.2	0.790123	0.158025
6	0. 0.774597	−119.8	0.493827	−59.1605
7	0.774597 −0.774597	208.178	0.308642	64.2526
8	0.774597 0.	444.818	0.493827	219.663
9	0.774597 0.774597	208.178	0.308642	64.2526

$I \approx -409.867$

By direct evaluation it can easily be verified that the exact value of the integral is as follows.

```
Integrate[f, {s, −1, 1}, {t, −1, 1}]
```

```
−437.295
```

Thus we need to further increase the number of integration points to get more accurate solution.

三维高斯积分：

Quadrature	Points	r_i	s_j	t_k	$w_i \times w_j \times w_k$
1x1x1	1	0.	0.	0.	8.
2x2x2	1	-0.57735	-0.57735	-0.57735	1.
	2	-0.57735	-0.57735	0.57735	1.
	3	-0.57735	0.57735	-0.57735	1.
	4	-0.57735	0.57735	0.57735	1.
	5	0.57735	-0.57735	-0.57735	1.
	6	0.57735	-0.57735	0.57735	1.
	7	0.57735	0.57735	-0.57735	1.
	8	0.57735	0.57735	0.57735	1.
3x3x3	1	-0.774597	-0.774597	-0.774597	0.171468
	2	-0.774597	-0.774597	0.	0.274348
	3	-0.774597	-0.774597	0.774597	0.171468
	4	-0.774597	0.	-0.774597	0.274348
	5	-0.774597	0.	0.	0.438957
	6	-0.774597	0.	0.774597	0.274348
	7	-0.774597	0.774597	-0.774597	0.171468
	8	-0.774597	0.774597	0.	0.274348
	9	-0.774597	0.774597	0.774597	0.171468
	10	0.	-0.774597	-0.774597	0.274348
	11	0.	-0.774597	0.	0.438957
	12	0.	-0.774597	0.774597	0.274348
	13	0.	0.	-0.774597	0.438957
	14	0.	0.	0.	0.702332
	15	0.	0.	0.774597	0.438957
	16	0.	0.774597	-0.774597	0.274348
	17	0.	0.774597	0.	0.438957
	18	0.	0.774597	0.774597	0.274348
	19	0.774597	-0.774597	-0.774597	0.171468
	20	0.774597	-0.774597	0.	0.274348
	21	0.774597	-0.774597	0.774597	0.171468
	22	0.774597	0.	-0.774597	0.274348
	23	0.774597	0.	0.	0.438957
	24	0.774597	0.	0.774597	0.274348
	25	0.774597	0.774597	-0.774597	0.171468
	26	0.774597	0.774597	0.	0.274348
	27	0.774597	0.774597	0.774597	0.171468